Wind, slope, and the want of a tail: a computational feasibility study of Eilmer of Malmesbury's flight

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Abstract

William of Malmesbury records that early in the eleventh century Eilmer, a fellow monk, fixed wings to his hands and feet, launched from a tower, travelled "more than a furlong" (about 200 m), and fell, breaking both legs. He blamed the missing tail. This paper tests the account with undergraduate mechanics (glide, slope lift, pitch stability, impact), public terrain and wind data, and a Monte Carlo sample over the unknown apparatus and launch day. From a walking-pace launch a glider loses about 11.5 m reaching its 15.1 m s^{-1} flying speed, so a long flight needs a strong wind. Flown into that wind over the valley, none of 40 000 sampled cases covers 200 m. Along the southwest-facing slope, in ridge lift, 6.9 % do (95 % confidence interval 6.6–7.1 %), at a median launch wind of 7.6 m s^{-1}. Without a tail a pitch disturbance doubles in a median 0.29 s, comparable to human reaction time, in a flight of about 24 s; a tail removes the divergence. A controlled arrival does not break the legs; one after loss of pitch control often does. The account is physically possible but marginal, and the missing tail is a sufficient cause of the fall. Each step is a student exercise with worked answers.

1 Introduction

William of Malmesbury, writing his *Gesta Regum Anglorum* in about 1125, records that Eilmer, an older monk of his own abbey, had in his youth fixed wings to his hands and feet, launched from a tower, and travelled "more than a furlong" before he fell, broke both legs, and was lame for the rest of his life [1]. Eilmer, William adds, used to say that he fell because he had forgotten to fit a tail. Lynn White's study of the passage established it as the earliest credible account of a gliding attempt in Western Europe [2].

The short account names a launch point, a distance, a wind, a failure, an injury, and a cause, and undergraduate physics can check each one. Could Eilmer's flight have happened at Malmesbury, and is a missing tail enough to explain the fall? No source describes Eilmer's apparatus beyond its wings, so the models sample its unknown parameters.

As a teaching case for undergraduate mechanics, the problem assumes calculus, Newtonian mechanics, and elementary probability, and suits students from the second year onwards and their instructors. Answering it needs the energy balance of a glider at launch, flight in a moving air mass, lift over a slope, the neutral-point criterion for pitch stability, the growth rate of an unstable mode, Monte Carlo estimation with confidence intervals, and the mechanics of a feet-first impact. Each step uses public data and a few lines of code, gives a number to check against the text, and becomes an exercise in Section 6.

2 The source and the site

William wrote a century after the flight and could have heard it from men who knew Eilmer [2, 3]. Supplementary S1 gives his Latin and a standard translation. He says Eilmer "collected the breeze on the summit of a tower" and that the violence of the wind and the swirling of the air brought him down. The traditional date is about 1000–1010. Aitcheson [4] argues it could be as late as the early 1040s. Launch weather is sampled from a modern climatology, so the year does not matter.

Other early flight accounts are mostly legend [5]. The only one with a stated device is that of Abbas ibn Firnas of ninth-century Córdoba, whom later Andalusian chronicles credit with a structure of feathers and silk and a bad landing, also blamed on a missing tail [2, 6]. Hallion [7] reads it as closer to a parachute descent. Neither gives dimensions or construction. With no historical basis for a specific design, the models sample wing area, mass, shape, and drag over wide ranges (Supplementary Table S1).



Figure 1: Malmesbury Abbey seen from the southwest. The view looks northeast from the Avon water meadows, about 300 m west-southwest of the abbey, up the slope the prevailing wind strikes. The present church is twelfth-century; Eilmer's Anglo-Saxon church and tower do not survive, so the launch point is taken as the abbey site and the tower height is sampled. Photograph: Dr Duncan Pepper (2012, Geograph and Wikimedia Commons), CC BY-SA 2.0.

The abbey stands on a narrow spur between two arms of the Avon. The sampled tower height, 15–25 m, brackets White's estimate of about 18 m [2]. Woosnam [8] puts the flight down the steep slope southwest of the abbey towards the river, the side shown in Fig. 1.

3 Methods

3.1 Terrain

The terrain comes from the Amazon Web Services (AWS) Terrain Tiles, a digital elevation model (DEM) with a grid spacing of about 6 m. Slope and aspect (the direction a slope faces)

are computed within 400 m of the abbey, along with a southwest–northeast elevation section. The grid smooths the steep edge of the spur, so slopes are lower bounds.

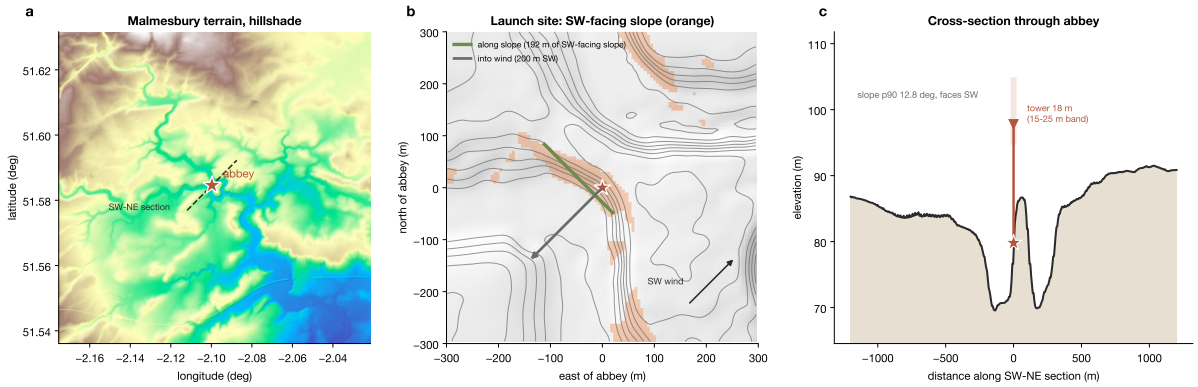


Figure 2: Terrain at Malmesbury Abbey. (a) Hillshaded DEM (AWS Terrain Tiles); the dashed line is the section in (c). **(b)** Southwest-facing ground steeper than 5° within 300 m of the abbey (star), in orange. The green line is the track along the slope (192 m); the grey arrow is the first 200 m of the track into the wind. **(c)** Elevation along the section, with the sampled tower heights.

3.2 Wind climatology

Hourly 10 m wind at the abbey for 2004–2023 (175 320 hours) comes from the ERA5 reanalysis of the European Centre for Medium-Range Weather Forecasts [9], through the Open-Meteo archive [10]. An hour is favourable when the wind blows from the southwest sector (191°–259°, centred on the slope aspect) at 5–12 m s^{-1}. Annual and seasonal fractions have 95 % confidence intervals (CI) across the twenty years. Modern data stand in for the eleventh century. Reconstructions of the North Atlantic Oscillation (NAO), the pressure pattern that sets the strength of the westerlies over Britain, disagree on its medieval phase [11, 12]. Neither indicates a long negative phase with weak westerlies.

3.3 Ridge lift

Wind blowing up a slope is deflected upwards, with a vertical velocity that to first order follows the surface gradient and decays with height,

$$w(x, z) = U \frac{\mathrm{d}h}{\mathrm{d}x} e^{-kz}, \tag{1}$$

where U is the wind speed, $h(x)$ the terrain height along the wind, z the height above the surface, and $k = 2\pi/L$ with L the horizontal scale of the slope. Supplementary Fig. S1 applies Eq. (1) to the measured section; the glide model uses the simpler surface estimate $w = \eta\, U \sin\theta_s$, where θ_s is the slope angle and $\eta \leq 1$ allows for flow that does not follow the surface. This lift extends 40 m on the track into the wind and 192 m on the track along the slope (Fig. 2b).

3.4 Glide performance

The glide ratio L/D of an unknown wing follows from the parabolic drag polar $C_D = C_{D0} + C_L^2/(\pi e A)$, whose maximum is

$$(L/D)_{\max} = \frac{1}{2}\sqrt{\frac{\pi e A}{C_{D0}}}, \tag{2}$$

where A is the aspect ratio (span squared over area), e the span efficiency, and C_{D0} the zero-lift drag coefficient [13]. An aspect ratio of 3–5 gives a span of one to two arm spans for areas of 5–10 m^2. Span efficiencies of 0.6–0.8 cover untwisted, non-elliptical wings [13]. The zero-lift drag $C_{D0} = 0.05$–0.15 adds the profile drag of a cambered membrane at a Reynolds number near 10^6 [14, 15] to the drag of the pilot's body, frame and bracing [16]. These ranges give $(L/D)_{\max} = 3.1$–7.9, with 4.7 at their centres. Every wing is flown at its best glide ratio, which favours the account.

A wing of area S carrying mass m at lift coefficient C_L trims at $V = \sqrt{2mg/(\rho S C_L)}$, 15.1 m s^{-1} for the nominal configuration of Fig. 3 (80 kg, 7 m^2, $C_L = 0.8$).

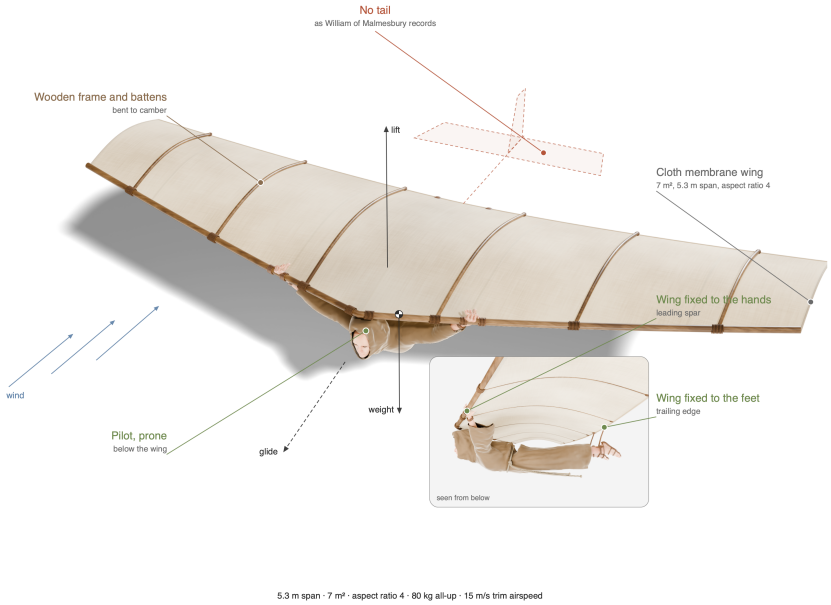


Figure 3: Nominal model configuration, not a reconstruction. A tailless cloth wing fixed to the prone pilot's hands (leading edge) and feet (trailing edge). Only the attachment and the missing tail come from the source; the dimensions are the centres of the sampled ranges (Supplementary Table S1). The dashed outline marks where a tail would be; arrows show lift, weight, glide and headwind.

3.5 Launch and glide trajectory

A man stepping off a tower leaves at about 2 m s^{-1} relative to the ground and dives until the airspeed reaches V, so the flight is integrated as a point mass at constant C_L,

$$\dot{V} = g\sin\gamma - \frac{D}{m}, \qquad \dot{\gamma} = \frac{1}{V}\left(g\cos\gamma - \frac{L}{m}\right), \qquad \dot{z} = -V\sin\gamma + w, \tag{3}$$

where V is now the airspeed, γ the descent angle below the horizontal, $L = \tfrac{1}{2}\rho V^2 S C_L$, and $D = L/(L/D)$. The airspeed at release is the jump speed plus the headwind.

Two tracks bracket the ground covered. On the first, the pilot heads southwest into the wind, down the slope towards the river [8], at ground speed $V\cos\gamma - U$ over the measured section. On the second, he flies northwest along the slope with the wind from the side, as hang-glider and sailplane pilots use ridge lift [17]. He crabs, heading partly into the wind to hold the track, at ground speed $\sqrt{(V\cos\gamma)^2 - U^2}$ over a contour 5.1 m below the abbey. William's "more than a furlong" becomes a ground range of at least 200 m.

3.6 Pitch stability

A wing is stable in pitch if the centre of gravity (CG) lies ahead of the neutral point, about which the pitching moment does not change with angle of attack. Both are measured from the leading edge as fractions of the mean chord c, x_{cg} and x_{np}. A cambered wing without a tail has $x_{\mathrm{np}} \approx 0.25$; a tail moves it aft, to 0.53 for the nominal tail, and adds pitch damping [13].

The pilot's mass centre lies between his hands, at the leading edge, and feet, at the trailing edge. Standing, it is at about 55 % of his height above the soles [18], and his reach from soles to raised hands is about 1.25 times his height. Laid out, it lies about 0.55 of the hand-to-foot distance behind the hands. The sampled range $x_{\mathrm{cg}} = 0.35$–0.55 runs from this straight posture forward to hips drawn up under the leading edge. A wider sweep, $x_{\mathrm{cg}} = 0.20$–0.60, shows where the conclusion would change.

With the CG aft of the neutral point, the short-period motion, linearised in angle of attack and pitch rate [13], has a positive real eigenvalue λ, so a disturbance doubles in $t_2 = \ln 2/\lambda$. t_2 is compared with the flight time and a human operator's effective time delay in manual control, about 0.15–0.3 s [19]. A nonlinear run tests an 8° gust, with control lost beyond 60° of pitch.

3.7 Monte Carlo sample

Because the apparatus and the day are unknown, 40 000 cases are drawn with a fixed seed from the uniform ranges of every parameter above (Supplementary Table S1), except the glide ratio, which follows from Eq. (2). The launch wind is drawn from the ERA5 southwest-sector hours at 3–16 m s^{-1}, assuming Eilmer chose a windy day. Each case is flown on both tracks with Eq. (3) and scored on whether it covers 200 m and whether t_2 is shorter than the flight, with and without a tail. Fractions have Wilson score 95 % CIs.

3.8 Impact

William records that Eilmer broke both legs and survived. In a feet-first landing the body stops over a distance d of leg flexion and ground give, so a vertical arrival speed v gives a mean force

$$F = \frac{mv^2}{2d}, \qquad \text{shared between two legs.} \tag{4}$$

The tibia breaks under axial and combined loads of 7.5–11.3 kN per leg [20–24], which sets a fracture-onset speed $v^\star = \sqrt{2dF_{\mathrm{tot}}/m}$ for $d = 10$–35 cm. Two arrivals are compared for cases that reach 200 m. A controlled glide without a flare meets the ground at the vertical speed from Eq. (3). After a loss of pitch control the craft strikes at airspeed V along a path 35°–70° below horizontal, with vertical speed $V \sin\theta$. Because Eq. (4) uses the mean force, peak loads are higher and the onset speeds are upper bounds.

4 Results

4.1 Site and wind

The abbey stands 10.2 m above the valley floor on a spur facing southwest (aspect 230°), sloping at a mean 5.2° and a 90th percentile of 12.8° within 400 m (Fig. 2). This southwest-facing ground runs 192 m northwest from the abbey but only 40 m southwest to the valley floor.

Over twenty years, 15.6 % of hours have southwest winds of 5–12 m s^{-1} (95 % CI 14.3–17.0 %), as do 21.1 % of winter hours (Fig. 4). Southwest winds of 8 m s^{-1} or more blow 296 hours a year (3.4 %, CI 2.9–3.9 %).

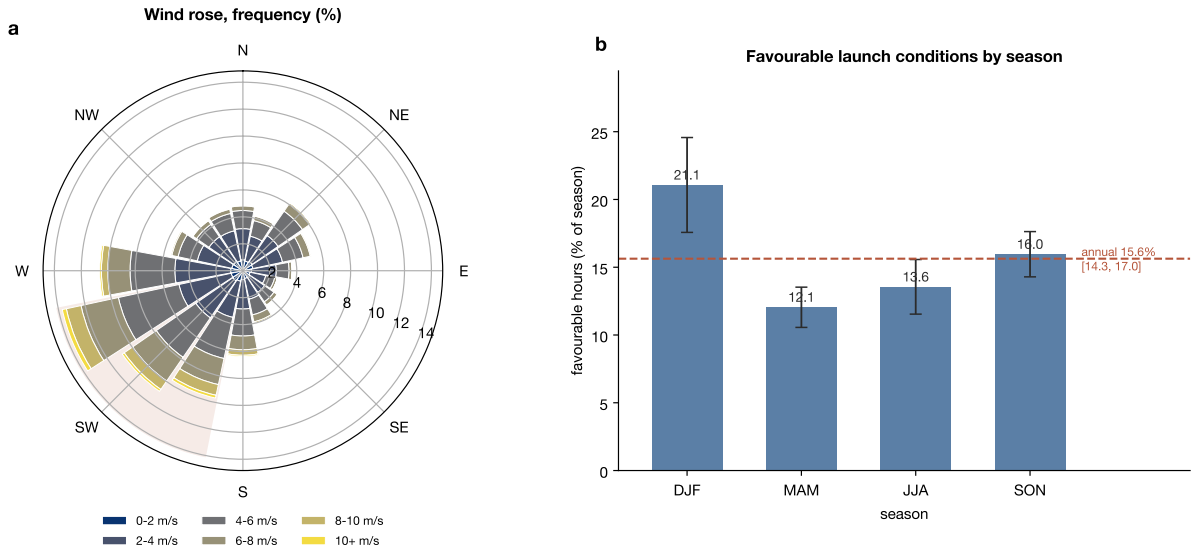


Figure 4: Wind at the abbey, 2004–2023. (a) Wind rose of ERA5 hourly 10 m wind (175 320 hours). **(b)** Fraction of hours from the southwest sector (191°–259°) at 5–12 m s^{-1}, for the year and by season (DJF is December–February), with 95 % confidence intervals across the twenty years.

In an 8 m s^{-1} wind the linear model (Eq. (1)) gives a peak updraught of 1.5 m s^{-1} southwest of the abbey (Supplementary Fig. S1), close to the surface estimate $U \sin\theta_s = 1.8$ m s^{-1} at the 90th-percentile slope. At its best glide the nominal craft sinks through still air at $V/\sqrt{1+(L/D)^2}$: 1.9 m s^{-1} at L/D 7.9 and 3.1 m s^{-1} at L/D 4.7. A strong wind can therefore hold up the better wings but not the poorer ones.

4.2 Launch and glide

Even a steady glide from the 18 m tower in still air would need L/D 11.1 to cover 200 m, beyond the range of Eq. (2). Eilmer starts from a jump, which is worse. Diving from 2 m s^{-1} to 15.1 m s^{-1} costs $(V^2 - v_0^2)/2g \approx 11.5$ m of height, and the best wing (L/D 7.9) lands 14 m out instead of the 142 m of a steady glide (Fig. 5b). A headwind supplies airspeed at release without the dive.

Flying into the wind, the same headwind slows him over the ground and the lift ends at the foot of the slope. No wing in Fig. 5a reaches 200 m in winds up to 14 m s^{-1}, and the longest flight is 110 m (dashed). Along the slope a crabbing craft can stay in the lift. There the best wing reaches 200 m above 5.7 m s^{-1} and the central wing above 10.6 m s^{-1}; the poorest never does (solid). Lowering the tower from 25 m to 15 m raises the threshold for the best wing from 3.8 m s^{-1} to 6.6 m s^{-1}.

4.3 Pitch stability

Without a tail the neutral point lies at 0.25 c, ahead of every sampled CG, so the wing is statically unstable in pitch (Fig. 6). Pitch damping holds off divergence until the CG is aft of 0.30 c. Across the sampled CG range a pitch disturbance doubles in 0.60 s at 0.35 c and 0.16 s at 0.55 c (0.12–0.89 s for half or double the pitch inertia); a 200 m flight lasts about 16 s. After an 8° gust the tailless craft passes 60° of pitch in 0.74 s; with a tail the gust dies away.

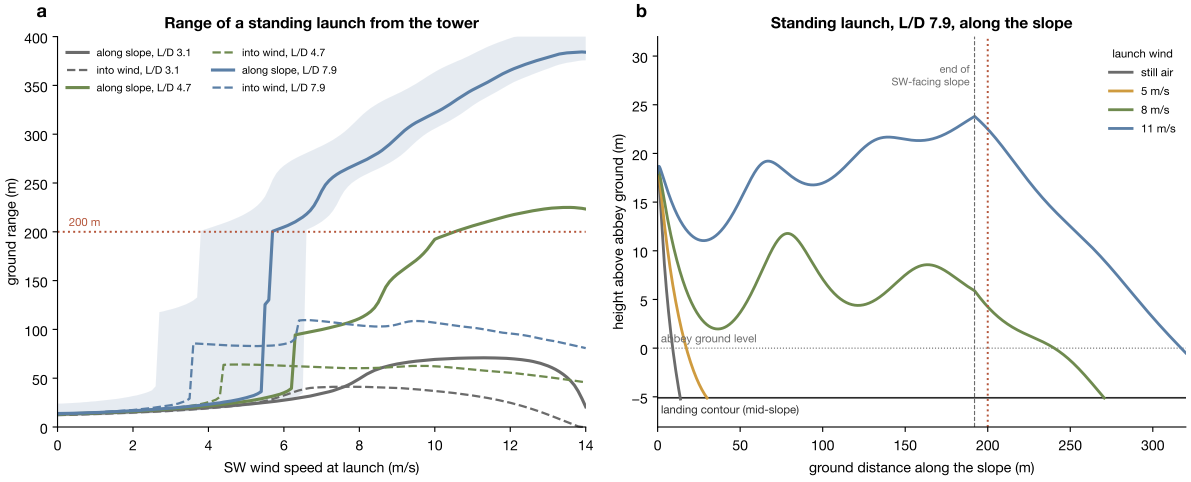


Figure 5: Range of a standing launch from the tower. **(a)** Ground range against launch wind for the nominal craft (Fig. 3) from the 18 m tower at the lowest, central, and highest glide ratios, along the slope (solid) and into the wind (dashed). The shaded band spans towers of 15–25 m for the best wing along the slope; the step near 5.5 m s^{-1} is where ridge lift first matches its sink. **(b)** Trajectories of the best wing (L/D 7.9) along the slope. In still air it dives to gain speed; in wind it rises and falls in the lift until the slope ends, then descends.

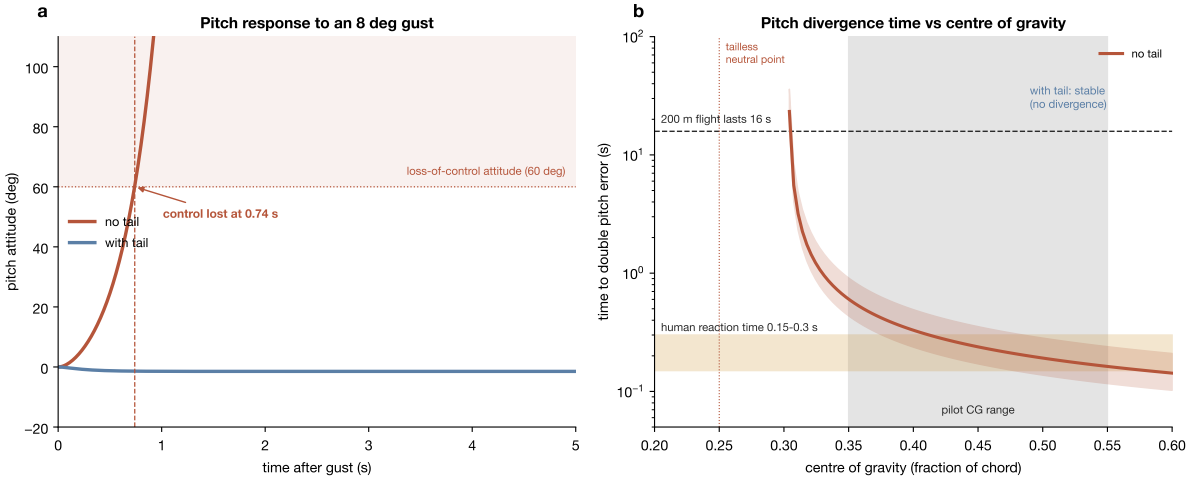


Figure 6: Pitch stability with and without a tail. **(a)** Pitch attitude after an 8° gust, CG at mid-range. **(b)** Doubling time of a pitch disturbance against CG position without a tail; the band spans half to twice the nominal pitch inertia. The grey band is the sampled CG range and the tan band the effective time delay of a human operator [19].

4.4 Monte Carlo sample

Into the wind none of the 40 000 cases reaches 200 m (95 % upper bound 0.01 %; Fig. 7a). Along the slope 2 748 cases do, 6.9 % (CI 6.6–7.1 %), against a median range of 83 m. They share a strong wind and a good wing: a median launch wind of 7.6 m s^{-1} (90 % above 5.4 m s^{-1}) and a median glide ratio of 5.8, against 4.7 over all cases. In these flights the tailless doubling time has a median of 0.29 s (99th percentile 1.6 s), shorter than the flight in every case (median 24 s; Fig. 7b). With a tail none diverges.

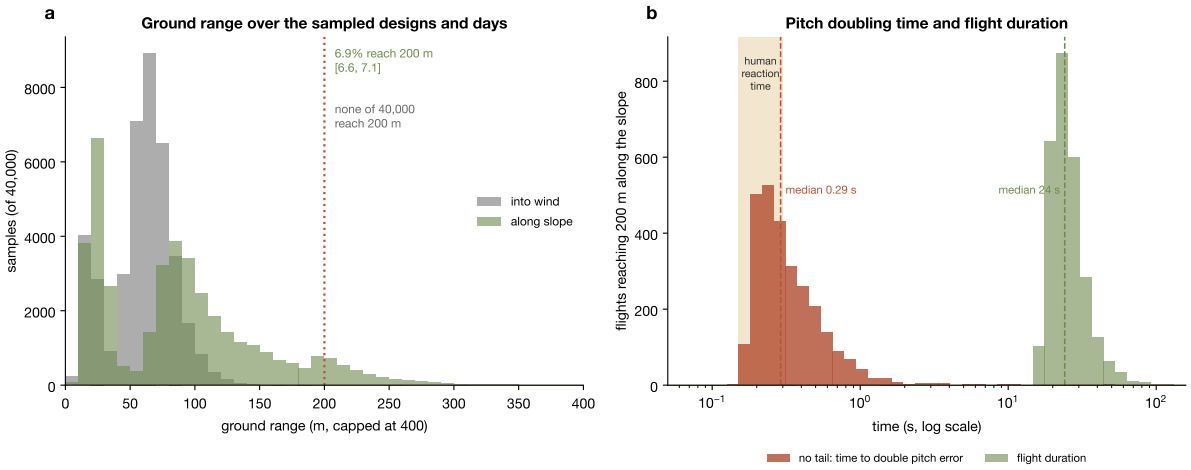


Figure 7: Monte Carlo sample of 40 000 cases. Apparatus parameters are drawn from Supplementary Table S1 and launch winds from the ERA5 southwest-sector hours. **(a)** Ground range on the two tracks. **(b)** Tailless doubling time and flight duration for the cases that reach 200 m along the slope; the tan band is as in Fig. 6.

4.5 Impact

With a stopping distance $d = 20$ cm the legs break above about 9.7 m s^{-1} of vertical speed, and across stopping distances and fracture loads the onset runs from 6.1 m s^{-1} to 14.1 m s^{-1} (Fig. 8a). A controlled glide that meets the ground without a flare arrives at a median 2.2 m s^{-1} and at most 4.1 m s^{-1}, below the lowest onset, so a missed flare does not break the legs (Fig. 8b). After a loss of pitch control, at airspeeds of 10.3–19.4 m s^{-1}, the median vertical speed is 10.1 m s^{-1} and 58 % of arrivals exceed the central onset. The median deceleration over 20 cm, about 26 g, is commonly survived in feet-first falls.

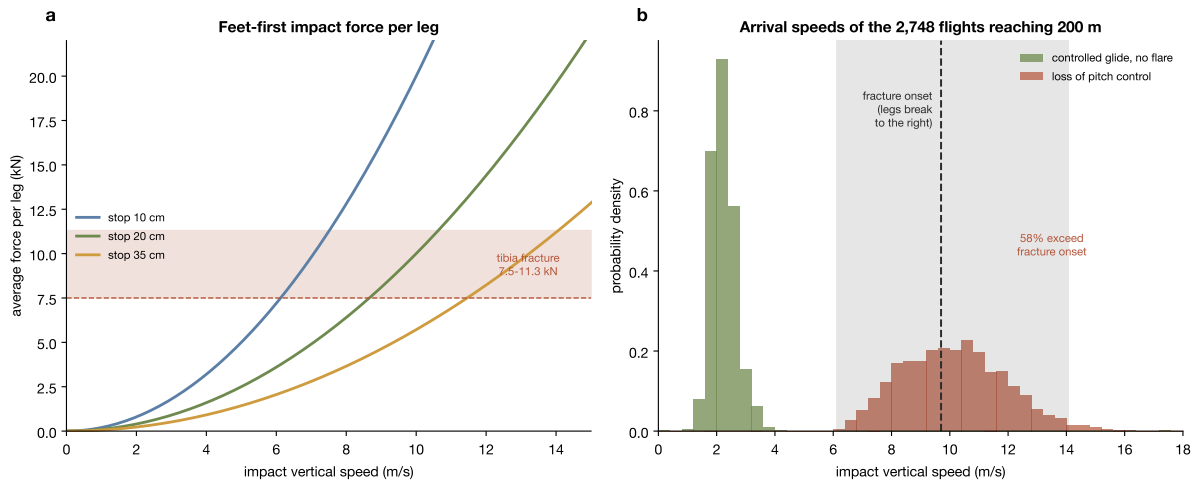


Figure 8: Arrival speed and leg fracture. (a) Mean feet-first impact force per leg against vertical speed for stopping distances of 10–35 cm, with the tibia fracture range (7.5–11.3 kN per leg). **(b)** Vertical arrival speeds of the cases that reach 200 m, gliding in without a flare and after loss of pitch control ($V \sin\theta$, $\theta = 35°$–$70°$). The grey band spans the fracture-onset speeds, with its centre dashed.

5 Discussion

5.1 What the account requires

A furlong from a tower of this height needs a strong southwest wind, a wing near the better end of what cloth and timber allow, and a track that stays over the slope. The headwind that gives Eilmer his launch speed also holds him back over the ground, so on the straight line out over the valley that Woosnam [8] proposed he runs out of slope first. Winds of $8\,\mathrm{m\,s^{-1}}$ or more blow about 296 hours a year, so a man who waited for one would not wait long, and William's phrase that Eilmer "collected the breeze" fits a launch timed to such a wind.

Without a tail the craft diverges in pitch within a fraction of a second, close to human reaction time. By shifting his body Eilmer could hold the attitude for a while, so tens of seconds of flight and then a loss of control fit the model. William names the trigger, the violence of the wind and the swirling of the air, and a gust that demanded a faster correction than he could make would end the flight in under a second (Fig. 6a). The models show only that he was unlikely to keep control, and that a tail would have removed the need.

The broken legs are consistent with a loss of control but do not prove one. Gliding in without a flare is too slow to break them in this model. The injury needs a fast, steep arrival, which a loss of pitch control gives, but so would a stall at height or a dive unrelated to the tail.

5.2 Which results depend on the assumptions

Three results rest on measurement or simple bounds, not on the sampled ranges. The slope faces the prevailing wind, and strong southwest winds are measured (Figs. 2 and 4). The launch dive costs $(V^2 - v_0^2)/2g$ of height whatever the wing, which rules out a still-air furlong from this tower. A tailless wing is unstable in pitch whenever the CG is aft of about $0.30\,c$, and a man fixed to it at his hands and feet has his CG near mid-chord. This would fail only if his mass centre lay ahead of $0.30\,c$, near his hands (Fig. 6b).

The fraction of cases reaching 200 m does depend on the ranges in Supplementary Table S1, and should be read as the share of the sampled ranges that works, not as a probability that the flight happened. It is most sensitive to the glide ratio and the ridge lift. The smoothed 6 m

terrain grid understates the real slope, which would raise the lift, while turbulence and separation behind the spur edge would lower it. The point-mass model holds C_L fixed, so the pilot cannot trade height for speed better than a stable glider, and the pitch analysis is longitudinal only. A large-eddy simulation over laser-scanned terrain and a six-degree-of-freedom model with a pilot would narrow these margins.

5.3 Limits of the comparison

The comparison assumes that a furlong means about 200 m of ground, that the tower stood at or near the abbey, and that William reports what Eilmer said. A shorter flight would be easier to explain, and a tower on lower ground harder. No model here tests whether the structure could carry the flight loads.

6 Using the study in teaching

Each exercise uses one topic from first- or second-year mechanics and one public data set or short script. They can be set singly, or together as a project in which students rebuild a figure and change one assumption. Worked answers are in Supplementary S4.

1. **Energy at launch.** A glider leaves a tower of height H at $v_0 = 2\,\mathrm{m\,s^{-1}}$ and needs $V = 15.1\,\mathrm{m\,s^{-1}}$ to fly. Ignoring drag, find the height lost in reaching V. How large a headwind makes the launch dive unnecessary? Integrate Eq. (3) in still air and compare the range with the steady glide estimate $H\,(L/D)$.

2. **Glide ratio from a drag polar.** Derive Eq. (2) by minimising C_D/C_L. Evaluate it at the corners of the ranges in Supplementary Table S1. Which parameter does the glide ratio depend on most, and which measurement on a model wing would narrow it?

3. **Flight in a moving air mass.** Draw the velocity triangle for a craft of airspeed V heading into a wind U, and for one crabbing along a track perpendicular to it. Show that the ground speeds are $V - U$ and $\sqrt{V^2 - U^2}$. Explain why a headwind helps the launch and shortens the range on the first track.

4. **Ridge lift and sink.** The upward air speed over a slope of angle θ_s is about $U \sin\theta_s$. The still-air sink rate is $V/\sqrt{1 + (L/D)^2}$. Find the wind at which the two are equal for L/D 4.7 and L/D 7.9 on a slope of 12.8°, and compare them with the ERA5 wind distribution.

5. **Static and dynamic pitch stability.** Write the two-state short-period model in angle of attack and pitch rate. Show that a positive real eigenvalue appears when the CG moves aft of a point that depends on the pitch damping, and compute the doubling time for the nominal craft at $x_{\mathrm{cg}} = 0.45$. Compare it with your reaction time from a ruler-drop test.

6. **Monte Carlo estimation.** Rerun the Monte Carlo with a different seed and sample size. Compute the Wilson interval for 2 748 successes in 40 000 trials by hand. Change one range in Table S1 and report how the reach fraction moves.

7. **Impact.** From Eq. (4), find the vertical speed that breaks a tibia for stopping distances of 10–35 cm. Estimate the vertical speed of a craft flying at V along a path 50° below horizontal, and of one that glides into the ground at its sink rate. What does each tell you about how Eilmer landed?

In exercises 2 and 6 students bound quantities nobody measured from physical arguments, sample them, and find which conclusions change when a bound moves.

7 Conclusion

A furlong's flight from the abbey tower is physically possible for a cloth wing, but only in a strong southwest wind, with a wing near the better end of what the materials allow, and on a track that follows the slope. Without a tail the craft is unstable in pitch. A disturbance doubles in about the time a man takes to react, so a flight of some tens of seconds ending in a loss of control in a gust fits the models; a tail removes the instability. The broken legs need a fast, steep arrival, which a loss of control provides. Eilmer's own explanation of his fall agrees with the physics, which cannot show that it was the only cause.

Data and code availability

The analysis code, data manifests, and figure scripts reproduce every number and figure in this paper. They are openly available at

`https://github.com/silverdavi/eilmer-flight-feasibility`
`https://doi.org/10.5281/zenodo.20706321`

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